A Comparative Analysis of SAP Implementation: Case Studies in Success and Failure at Siemens Healthineers and Lidl

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Introduction

"Think about the core processes needed to run a company: finance, human resources, manufacturing, procurement, supply chain, sales, and more. Enterprise resource planning, or ERP, helps you manage all those processes in a single, integrated system" (*What is SAP ERP? key features & benefits | SAP*). ERP systems allow companies to utilize a single system to do the work that multiple systems did before. ERP systems are cross-functional, process-centered and create a single, common database (Setterstrom, 2025). An ERP system can provide many benefits to a business including but not limited to: delivering better customer satisfaction, improved vendor performance, improved resource utility, improved information accuracy, and improved decision-making capabilities (*10 top benefits of implementing an ERP system*). Migrating from legacy systems to ERP software can be a huge undertaking, so many organizations bring in experts to help run a smooth implementation plan (Hernandez, 2025). One such group of experts is SAP. "SAP ERP is a comprehensive software system that streamlines processes, improves productivity, and provides real-time insights across your entire organization" (*What is SAP ERP? key features & benefits | SAP*). Even with bringing in experts, studies show that approximately 50-75% of ERP projects fail to meet their objectives (Bradley, 2019). This paper will discuss a successful SAP implementation in Siemens Healthineers and an unsuccessful SAP implementation in Lidl.

Successful Implementation of SAP at Siemens Healthineers

Introduction to Siemens Healthineers AG

Siemens Healthineers AG is a leading global medical technology company originating from Siemens AG's healthcare division. In 2016, Siemens spun off its medical technology operations to "secure the financial strength and entrepreneurial flexibility needed to continue pioneering breakthroughs in healthcare" (*About Siemens Healthineers*). The company completed its initial public offering on the Frankfurt Stock Exchange in April 2018 and was elevated to the DAX in 2021. In fiscal year 2024, results showed approximately €22.4 billion in revenue with a workforce of around 72,000 employees worldwide (*Siemens healthineers achieves targets for 2024 fiscal year with strong fourth quarter* 2024). The business operates in three segments: imaging systems, laboratory diagnostics, and point-of-care testing solutions, serving healthcare providers across more than 180 countries (*Siemens healthineers achieves targets for 2024 fiscal year with strong fourth quarter* 2024).

Following the April 2018 spinoff, Siemens Healthineers initiated project P58 to build an independent ERP system harmonizing sales, service, and finance processes across Regional Companies. This initiative was essential for establishing operational autonomy from Siemens AG's IT infrastructure. The project involved deploying SAP templates across

more than 60 countries while simultaneously upgrading to the S/4HANA platform. The company's values of 'learn together, step up boldly, and win together' guided this implementation (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

Strategic Context for SAP Implementation

Phase 1: Initial SAP Adoption (2018 Spinoff)

Legal and structural requirements necessitated a new ERP system during the carve-out, as the company's IT systems were integrated with Siemens Group's infrastructure (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*) (*Go for the Siemens' healthineers placing* 2018). A standalone ERP system would enable customized processes meeting healthcare sector requirements (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*). The company operated approximately 12 different legacy systems prior to reorganization (*Go for the Siemens' healthineers placing* 2018). Project P58 aimed to consolidate these into one global platform, migrating 70 countries into a single unified system (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*).

Following the spinoff, the initial work focused on establishing the foundation for the new system. The first step involved setting up an SAP template mapping all global sales, service and finance processes before rolling out the system across more than 60 countries. The SAP implementation addressed multiple strategic objectives: harmonizing and optimizing processes across countries, enabling standardization and digitalization, ensuring IFRS15 compliance, and improving transaction data transparency (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

System consolidation required investments of approximately €100 million. However, projected returns included annual cost savings of €40-50 million, with additional savings of €190 million expected after fiscal year 2018 and medium-term annual savings reaching €240 million (*Go for the Siemens' healthineers placing* 2018). As Global System Lead Sonja Fischer put it, "With the innovation this is saving the company a lot of money if you're doing it in one system instead of having innovations in 12 systems. So, we are saving time and money" (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*).

Phase 2: Parallel Deployment and S/4HANA Upgrade (2021-2023)

Prior to upgrading and deploying the new SAP templates in 2021, the company faced a critical strategic decision regarding sequencing of deployment and S/4HANA upgrade. At this juncture, Siemens Healthineers brought Deloitte onboard as implementation partner. Rather than pursuing sequential implementation, the experts at Deloitte and Siemens Healthineers decided on an unprecedented approach: executing both initiatives simultaneously, an approach never attempted at this scale (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*). As Fischer observed,

"This is quite unique in the SAP world, doing rollouts and upgrades and moving to RISE all at the same time" (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*).

The scale and complexity of the undertaking were substantial. Key parameters illustrated the project's magnitude: around 75 satellite systems, more than 275 interfaces, and over 750 level-3 processes. Managing this massive number of tasks, systems, resources, and data within the merged timeline required broad-based experience and expertise. The technical complexity of the ERP upgrade impacted the existing data migration strategy and required modification of the existing ETL (extract, transform, load) tools. With the many products, business models, and far-reaching regulations specific to the healthcare business, the project became even more complex (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

The S/4HANA platform was selected for its innovation capabilities. Global Rollout Leader Thomas Hauer emphasized: "We're very sure that with our global SAP system, we are right now implementing, we have the best basis to implement AI functions in the future" (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*). The experts at Deloitte and Siemens Healthineers decided to proceed in several waves to execute this ambitious project, essentially industrializing the deployment process by clustering country organizations according to their complexity and geography. Implementation proceeded in waves: 7 countries by October 2021 (Wave 1), S/4HANA go-live in October 2022, and 14 additional countries by January 2023 (Waves 2 and 3) (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*). As of early 2025, the system operated in 36 countries with ongoing rollout toward 70 countries (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*).

Phase 3: RISE with SAP Cloud Migration (October 2023)

In October 2023, Siemens Healthineers selected RISE with SAP to advance digital transformation. CIO Stefan Henkel stated: "This migration to the cloud is especially important for us as we continue to invest in technology to support our continued growth". The cloud platform would streamline data sources, improve performance and sustainability, and enhance process governance. SAP SE Executive Board member Thomas Saueressig noted: "Together, we will leverage the cloud's potential to drive innovation, reduce costs, improve agility" *(Siemens Healthineers picks rise with SAP to boost performance, 2023)*.

Implementation Success and Realized Benefits

With Deloitte as implementation partner, project P58 achieved success across all objectives, delivering the system to 30 countries with additional rollouts continuing. The Deloitte team mobilized a wide range of experts from its worldwide network to set up a fit-for-purpose project structure with cross-functional team leads and a global reach, utilizing a mix of onsite, nearshore, and offshore resources. The creation of an integrated staffing plan for both the deployment and the upgrade leveraged staff synergies, while Deloitte Managed

Services were and will continue to be used for support and DevOps (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

The implementation utilized a wave-based methodology with centralized configuration teams and local hub teams employing parallel and synchronized testing. The key to rapid rollout was industrializing the deployment strategy, with one central team for template configuration working with a series of local hub teams and synergy-building solutions such as parallel and synchronized testing. Critical success factors included the close, trust-based collaboration of participating teams and how effectively the project team managed the integration of Healthineers' various IT architectures (production landscape, project landscape, sandboxes), the S/4HANA upgrade sequences, the validation test cycles, and the system architecture linking production operations with project activities. Despite pandemic constraints and remote work requirements, as Stephan Bode, Partner and Offering Lead SAP at Deloitte, noted: "Precise planning was the key to the project, but also courage, determination, and trust. Particularly with the teams working mainly remotely during the pandemic, the single most critical success factor was probably 'having each other's back'. I am really proud to have been a part of this success story" (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

The S/4HANA upgrade reached its first go-live milestone in just twelve months. The project delivered results eight months faster than sequential implementation while remaining under budget (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*). Financial performance validated the business case: €100 million investment generated annual savings of €40-50 million, with additional savings of €190 million post-fiscal year 2018 and medium-term projections of €240 million annually (*Go for the Siemens' healthineers placing,* 2018). Fischer explained: "With the innovation this is saving the company a lot of money if you're doing it in one system instead of having innovations in 12 systems. So, we are saving time and money" (*Siemens Healthineers: Implementing one global sales and customer service system based on SAP*).

Operational benefits included enhanced transaction data transparency, optimized business processes, and productivity gains through regional synergies. The S/4HANA platform delivered flexible, customizable features enabling productivity gains through standardized roles and processes while creating potential for digital transformation and emerging technologies including artificial intelligence. A number of enhancements were developed and introduced while the upgrade was still in progress, and S/4HANA enables the team to deliver ongoing optimizations as they roll out the system across the production landscape through DevOps. Thanks to this agile approach, more than 1,100 small to medium customizations offered key benefits even for countries in the earlier rollout waves (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

The October 2023 RISE with SAP announcement represented the transformation's culmination and foundation for future innovation. Through strategic SAP collaboration, Siemens Healthineers deployed SAP S/4HANA Cloud private edition, SAP Business Technology Platform, and SAP Signavio solutions to streamline data sources and improve business process performance and sustainability. CIO Henkel emphasized: "By embracing SAP technology, we will be able to reduce complexity and highly standardize our processes while delivering on our promise to pioneer breakthroughs in healthcare. This migration to the

cloud is especially important for us as we continue to invest in technology to support our continued growth" (*Siemens Healthineers picks rise with SAP to boost performance,* 2023). The implementation exemplified organizational values of "learn together, step up boldly, and win together," establishing a model for digital transformation in healthcare technology. As Christiane Reich, Head of IT EMEA and Member of the P58 Steering Board, stated: "This was a great team effort that would not have been possible without the perseverance, passion and precise planning of everyone involved. I am extremely proud of the entire team and am looking forward to bringing the innovations made possible by S/4HANA to every country organization in the enterprise" (*SAP transformation: Deployment and S/4HANA upgrade in parallel: Deloitte Deutschland*).

Lidl's €500 Million SAP Debacle: Lessons in Enterprise System Failure

Introduction

In 2011, Lidl, the German discount retailer owned by Schwarz Group, launched a company-wide ERP initiative to replace its legacy inventory and merchandise-management systems "Wawi" (Dolfing, 2020) with SAP software (Baumann, 2024).

The initiative, called eLWIS electronic Lidl merchandise management and information system, was designed to unify processes across thousands of stores and logistics centres. The anticipated benefits included better inventory control, enhanced forecasting, faster reporting, and reduced operational inefficiencies (Dolfing, 2020).

However, despite its ambitious objectives and significant financial investment, Lidl's SAP implementation failed. After seven years of development and deployment, the company reverted to its previous system, reporting losses of more than €500 million (Baumann, 2024) (Kimberling, 2025). This paper explores the reasons behind this failure and its implications for ERP projects globally.

Background

Lidl operates over 10,000 stores across Europe and the United States, offering high-quality goods at low prices. Its operational model relies on efficient inventory management, streamlined logistics, and strict cost control (Dolfing, 2020).

By 2011, Lidl recognized the need to modernize its IT infrastructure to maintain competitiveness. The eLWIS project aimed to standardize processes across stores and logistics centres, improve data visibility, and facilitate real-time decision-making. SAP software was selected due to its global reputation and comprehensive ERP capabilities (Baumann, 2024). About thousands of staff and hundreds of consultants implemented a new company wide system. The initial estimated cost was close to €80 billion (Dolfing, 2020).

Lidl went live with eLWIS in 2025 in stores and hubs in Austria, Ireland and some parts of US. By 2016 Lidl completed integrating over 30 systems from the existing infrastructure. Lidl won a award from SAP for being it's one of the best customers (Dolfing, 2020).

Despite careful planning, the project encountered significant challenges. Over seven years, it faced escalating costs, complex customizations, leadership turnover, and resistance to change. Ultimately, the initiative was abandoned in favour of the company's legacy system (Kimberling, 2025).

Implementation Strategy

Lidl selected SAP as its ERP platform and engaged external consultants to manage deployment. The company attempted a global rollout across multiple countries and thousands of stores. Internal teams worked alongside external experts to customize the software to Lidl's unique business processes (Baumann, 2024).

Root Causes of Failure

Resistance to Changing Business Processes

A core problem was Lidl's insistence on preserving its legacy business processes, particularly inventory valuation. Lidl valued inventory based on purchase prices, whereas SAP operates on retail price valuations (Baumann, 2024).

Rather than adapting internal processes to SAP best practices, Lidl customized the software to replicate existing methods. This approach created a massive gap between the system's design and Lidl's operational requirements (Dolfing, 2020).

Over-Customization

Lidl's customization of SAP was extensive. While customization initially appeared necessary, it introduced technical complexity and slowed deployment (Kimberling, 2025). Extensive modifications also increased maintenance costs and made future upgrades difficult, creating a long-term technical debt (Sumner, 2005).

Leadership Turnover and Strategic Misalignment

During the seven-year project, leadership at Lidl changed multiple times. Each transition resulted in shifting priorities, inconsistent project oversight, and unclear decision-making. The lack of a stable strategic vision undermined progress and coordination (Baumann, 2024).

Weak Change Management

ERP projects require careful change management. Lidl failed to adequately train staff and communicate the benefits of the new system. Employees often reverted to old habits, limiting the system's effectiveness (Baumann, 2024) (Bradley, 2019).

Overreliance on External Consultants

Although external consultants provided expertise, Lidl's internal team lacked the skills to oversee the project independently. The company relied heavily on consultants for critical decisions, diluting internal control and accountability (Kimberling, 2025).

Financial and Operational Impact

Lidl's abandoned SAP implementation had significant financial consequences. Reports estimate €500–600 million spent without achieving intended benefits (Baumann, 2024) (Kimberling, 2025). Operationally, the failure led to Continued reliance on legacy systems, delaying modernization. Disruption to store and logistics processes due to incomplete SAP deployment. Lost opportunity for data-driven decision-making and process efficiency. Beyond direct costs, the failure likely affected Lidl's reputation as an efficient and innovative retailer (Dolfing, 2020).

Lessons Learned

1. **Align processes with software best practices:** Organizations should adapt workflows to ERP capabilities rather than forcing software to fit outdated methods (Davenport, 1998).
2. **Limit customization:** Avoid extensive modifications that complicate maintenance and upgrades (Sumner, 2005).
3. **Stabilize leadership:** Executive commitment and strategic alignment are critical to ERP success (Scott and Vessey 2003).
4. **Invest in change management:** Train employees, communicate benefits, and support adoption (Bradley, 2019).
5. **Retain internal ownership:** External consultants can advise, but the organization must retain control (Kimberling, 2025).

These lessons highlight that ERP failure is rarely a purely technical problem; organizational culture, leadership, and people play a central role.

Conclusion

Reviewing these two case studies, it can be seen that there are various methodologies of conducting implementations of ERP systems, with multiple points in which success or failure can result. Thorough preparation, prework, and analysis is required as well as robust change management, risk identification/remediation, and ensuring that quality sponsors are on board. Senior leadership needs to champion these efforts, including ensuing that communication is in place and utilized to bring staff along on the journey. Case studies are effective in ensuring that lessons learned from other implementations with similarly sized and industry specific are taken into account and factor into project planning, risk mitigation/planning to set the program up for successful implementation.

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